United International University (UIU)

Dept. of Computer Science and Engineering (CSE)

Mid Exam Year: 2025 Trimester: Fall

Course: CSE 2215 Data Structure and Algorithms-I

Total Marks: 30, Time: 1 hour 30 minutes

(Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules)

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1. a) Simulate the **Selection Sort** algorithm to sort the following array in **descending** order, showing the state of the array after each iteration of the outer loop. [3]

   [30 40 10 32]

   b) Consider the following algorithm that finds the maximum and minimum values in a given 0-indexed array A of size n. Analyze the time complexities (best, worst, and average cases) of the algorithm and express them using asymptotic notations.: [4]

```
max=A[0];
min=A[0];
for(i=1; i<n; i++){
    if (A[i]>max)
      max=A[i];
    else if (A[i]<min)
      min=A[i];
}
```

2. a) For each of the following arrays, determine the total number of shifts required to sort the array in **descending** order using the **Insertion Sort** algorithm. [3]

   Data Set-I: 10, 20, 30, 40
   Data Set-II: 40, 30, 20, 10
   Data Set-III: 30, 10, 40, 20

   b) Find the memory location of A[60][70] if loc(A[15][20])=2000. Assume row-wise memory is allocated in the double type array A[80][100], where each data is 8 bytes. [2]

3. a) Find the number of comparisons required to locate the position of the character 'U' in the given string using **Binary Search** algorithm. Show the results of each comparison step-by-step. [4]

   Z X U R N H F D B A

   b) Write a recursive algorithm **printSeries(n)** to print all numbers from 1 to n except the numbers that are divisible by 3. [3]

   Examples:

Function Call	Output
printSeries(4)	1 2 4
printSeries(6)	1 2 4 5
printSeries(10)	1 2 4 5 7 8 10

Note: You are NOT allowed to use loops or any extra parameters in the recursion.

4. a) Suppose a doubly linked list is headed by a pointer head and contains four nodes with data values 70, 80, 90, and 100, respectively. Each node contains three fields: ***prev** (pointer to the previous node), **value** (data value), and ***next** (pointer to the next node).

   i) Draw a diagram representing the doubly linked list. [1]
   ii) Identify the pointer expressions for each node in the linked list with respect to the head. [2]
   iii) Write an algorithm to insert a new node with the value 85 between the nodes containing the values 80 and 90. [2]
   iv) Write an algorithm to delete the node containing the value 90. [2]

   b) Reverse the string "CSE" using a stack implemented with a linked list. For each push and pop operation, display the status of the stack. [2]

   c) For a queue of size m implemented using an array, explain how to check for overflow and underflow conditions. [2]